

BALANCE–FIELD FRAMEWORK (BFG)

The BFG Canonical Universal Reclosure Architecture

A Self-Contained Mathematical Whitepaper on Determinate Formation, Exact Neutral Resolution, Persistent Recursive Closure, Endogenous Reciprocal Order, and the Canonical Emergence of New Structure

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Scientific status. Independent theoretical preprint and falsifiable research program. Exact operator identities and theorem statements are mathematical consequences only under their stated assumptions. The paper establishes a self-contained canonical reclosure architecture on a declared admissible operator category. It does not claim that physical, biological, cognitive, or phenomenal systems have already been empirically shown to realize that category.

Abstract

This paper presents a single self-contained mathematical formulation of canonical universal reclosure in the Balance–Field Framework. The construction begins with determinate formation on a real Hilbert-space profile sector, eliminates neutral mediation exactly through the positive load $Y=K\dagger K$ and the complementary resolvent pair $C_N=(I+Y)^{-1}$ and $B_N=Y(I+Y)^{-1}$, selects recursively persistent structure through a bounded non-decay subspace in the graph metric $G=I+Y$, and derives a unique endogenous dual-order state from reciprocal balance of the two persistent closure loads. The resulting cross-fed packet is non-amplifying in the structural metric, its active support is transported by the canonical polar partial isometry, and the next geometry is rebuilt by a positive Gram construction. A universal formation gate then decides whether the reconstructed next-level formation operator possesses an isolated simple negative mode. When it does, the same quartic admissibility functional that generates the initial nontrivial branch produces exactly two polarity-related next-level minima. This yields a precise emergence mechanism: a new closure candidate is generated by the same operator grammar rather than by a stage-specific law. Genuine structural novelty is tested against the canonically inherited active-support reference of the predecessor, not against the predecessor merely as a whole; projection, support reduction, or recursion depth alone therefore cannot certify emergence. The architecture is totalized by an absorbing terminal object for failed generation and admits an exact compact representation through the neutral contrast operator $Z=(I-Y)(I+Y)^{-1}$. An intrinsic future-behavioral quotient removes target-dependent coarse-graining, while a no-choice reduction protocol converts residual ambiguity into explicit failure rather than post-hoc tuning. Independent numerical audits on 10,000 seeded complex random instances verify the neutral partition, reciprocal balance, compact/expanded equivalence, polar support identity, positivity preservation, and the analytic non-expansion bound to numerical precision. A separate 10,000-instance emergence audit, together with 1,000 commuting inheritance controls, tests the distinction between mere support inheritance and inequivalent next-formation structure. The final result is a compact but auditable mathematical preprint in which formation, persistence, reclosure, novelty, and failure are all explicitly defined.

Keywords: *Balance–Field Framework; universal reclosure; exact neutral resolution; positive operator geometry; recursive persistence; reciprocal balance; polar decomposition; structural emergence; canonical quotient; no-retuning.*

Contribution and consolidation note

This preprint consolidates a sequence of internal BFG reclosure developments into one canonical presentation. Developmental version labels are intentionally removed from the mathematical body. Only constructions that are mutually compatible in the source series are retained: Level-0 determinate formation, exact neutral elimination, positive master geometry, bounded non-decay persistence, graph-metric projection, reciprocal endogenous order, polar support transport, positive Gram rebuild, totalized failure, the compact neutral-contrast representation, the intrinsic behavioral quotient, and the freedom-closed reduction discipline. Earlier formulations in which a carrier-

specific scalar order coordinate entered the generator are not retained as generator primitives; such scalars appear only as optional downstream empirical readouts.

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Declarations

1. Scope, admissible claims, and closure criterion

The purpose of this paper is not to assign a different fundamental law to each physical, biological, or cognitive regime. The mathematical target is narrower and stronger: define one admissible operator category and one reclosure grammar such that every nonterminal iterate reconstructs the data required for the next iterate without importing a stage label, a sector-specific response law, a tunable persistence window, a free transport matrix, or a fitted order coordinate.

Mathematical closure and natural realization are therefore separated. Mathematical closure concerns whether the generator is a well-defined totalized self-map of its declared category. Natural realization concerns whether independently measured systems can be mapped into the resulting invariant classes without changing the generator. The first is a theorem-level question under explicit assumptions; the second remains empirical.

Table 1. Epistemic discipline used throughout the preprint.

Claim class	Meaning in this paper
Definition	Introduces a mathematical object, equivalence, or failure state.
Theorem	Exact consequence of explicitly stated assumptions.
Conditional theorem	Exact consequence once additional structural assumptions are admitted.
Construction	Canonical operator or map fixed by previously defined objects.
Numerical audit	Finite-precision check of identities or bounds; not empirical validation.
Empirical hypothesis	Proposed realization by a natural carrier; never used to define the generator.

Closure criterion: A stage transition is not counted as mathematically closed if it requires a new domain-specific fundamental response function, a post-hoc role change, a target-aware threshold, a selectable quotient, or a carrier map chosen only after the desired outcome is known.

2. Mathematical setting and admissible state category

Let $\mathcal{H}$ be a real or complex separable Hilbert space as appropriate to the operator sector under study. Determinate formation is stated first on a finite-dimensional real profile space, where the symmetry-breaking minimum can be proved without a continuous phase degeneracy. Neutral resolution and recursive closure are then formulated for bounded operators and extended to nonnegative self-adjoint quadratic forms.

An admissible nonterminal state is represented abstractly by the tuple

$$X = (D, \mathcal{D}, \mathfrak{c}, \mathfrak{N}, K, Y, R_C, \mathcal{W}, P, \omega, \mathcal{G}) \quad (1)$$

where D is the active difference/carrier object; $\mathcal{D}$, $\mathfrak{c}$, and $\mathfrak{N}$ are self-adjoint capacity operators for distinction, coherence, and neutral admissibility; K is the formation operator; Y is the nonnegative master load; R_C is the recursive closure operator; $\mathcal{W}$ is the persistent witness subspace; P its positive-metric orthogonal projector; ω the endogenous reciprocal-order pair; and $\mathcal{G}$ denotes the reconstructed positive geometry required for reclosure.

$$X_adm := X_adm \cup \{\perp\} \quad (2)$$

The symbol $\perp$ denotes terminal non-generation. It is not a hidden repair state: once reached, the universal map remains at $\perp$.

$$U(\perp) = \perp \quad (3)$$

The admissible category is intentionally strict. Degenerate determinate-formation minima, non-isolated persistence spectra, vanishing dual loads, or failed positivity do not trigger an alternative fitted rule; they trigger terminal failure.

3. Determinate formation from an isolated negative mode

3.1 Primitive trinitarian capacity structure

On a finite-dimensional real Hilbert space V , introduce three self-adjoint capacity operators:

$$\mathcal{T} = (\mathcal{D}, \mathfrak{c}, \mathbf{N}) \quad (4)$$

$$\mathbf{K} := \mathfrak{c} + \mathbf{N} - \mathcal{D} \quad (5)$$

The admissibility functional is the coercive quartic

$$\mathcal{A}_{\text{Adm}}(\delta) := \frac{1}{2}\langle \delta, \mathbf{K}\delta \rangle + \frac{1}{4}\|\delta\|^4, \quad \delta \in \mathbf{V}. \quad (6)$$

The quartic term bounds the functional below. The spectral sign of $\mathbf{K}$ determines whether the trivial state remains the unique minimum or whether a nonzero branch forms.

$$\lambda_0 := \min \sigma(\mathbf{K}), \quad \text{mult}(\lambda_0)=1, \quad \Delta := \lambda_1 - \lambda_0 > 0. \quad (7)$$

Theorem 1 — Determinate nontrivial formation. If $\lambda_0 < 0$ is a simple minimum eigenvalue of $\mathbf{K}$ with unit eigenvector $\mathbf{e}_0$ and positive spectral gap Δ , then $\mathcal{A}_{\text{Adm}}$ has exactly two global nonzero minima, related only by polarity:

$$\delta^*_{\pm} = \pm \sqrt{-\lambda_0} \mathbf{e}_0. \quad (8)$$

If $\lambda_0 \geq 0$, the unique minimum is $\delta=0$. If $\lambda_0 < 0$ but is degenerate, the minimum set is not a unique polarity pair and the strict determinate-formation gate fails.

Proof idea. Write $\delta = r\mathbf{u}$ with $\|\mathbf{u}\|=1$. Rayleigh–Ritz gives $\langle \mathbf{u}, \mathbf{K}\mathbf{u} \rangle \geq \lambda_0$. Hence $\mathcal{A}_{\text{Adm}}(r\mathbf{u}) \geq \frac{1}{2}\lambda_0 r^2 + \frac{1}{4}r^4$, with equality only in the lowest eigendirection. The scalar quartic is minimized at $r^2 = -\lambda_0$ when $\lambda_0 < 0$. Simplicity leaves only $\pm \mathbf{e}_0$.

3.2 Why this is a formation theorem rather than a naming rule

No physical or biological label appears in the theorem. The result only states when a previously unformed admissible profile sector develops a determinate nonzero configuration. This same criterion will be reapplied after every successful reclosure; therefore formation is not confined to an initial stage.

4. Exact neutral resolution and positive master geometry

4.1 Exact elimination of the neutral auxiliary variable

Let $\mathbf{A}$ map the active difference channel into a neutral mediation channel. Consider

$$\mathcal{A}_{\text{C}}(\mathbf{D}, \mathbf{v}) := \frac{1}{2}\|\mathbf{D} - \mathbf{A}^\dagger \mathbf{v}\|^2 + \frac{1}{2}\|\mathbf{v}\|^2. \quad (9)$$

$$\mathbf{Y} := \mathbf{A}^\dagger \mathbf{A} \geq 0. \quad (10)$$

$$\mathbf{v}^* = (\mathbf{I} + \mathbf{A}\mathbf{A}^\dagger)^{-1} \mathbf{A}\mathbf{D}. \quad (11)$$

Exact elimination yields the complementary bounded resolvents

$$\mathbf{C}_{\text{N}}(\mathbf{Y}) := (\mathbf{I} + \mathbf{Y})^{-1}, \quad (12)$$

$$\mathbf{B}_{\text{N}}(\mathbf{Y}) := \mathbf{Y}(\mathbf{I} + \mathbf{Y})^{-1} = \mathbf{I} - \mathbf{C}_{\text{N}}(\mathbf{Y}). \quad (13)$$

$$\mathbf{C}_{\text{N}}(\mathbf{Y}) + \mathbf{B}_{\text{N}}(\mathbf{Y}) = \mathbf{I}, \quad 0 < \mathbf{C}_{\text{N}} \leq \mathbf{I}, \quad 0 \leq \mathbf{B}_{\text{N}} < \mathbf{I}. \quad (14)$$

$$\mathcal{A}_{\text{C}}^{\wedge \{\min\}}(\mathbf{D}) = \frac{1}{2}\langle \mathbf{D}, \mathbf{C}_{\text{N}}(\mathbf{Y})\mathbf{D} \rangle. \quad (15)$$

$$\frac{1}{2}\|\mathbf{D}\|^2 = \frac{1}{2}\langle \mathbf{D}, \mathbf{C}_{\text{N}}\mathbf{D} \rangle + \frac{1}{2}\langle \mathbf{D}, \mathbf{B}_{\text{N}}\mathbf{D} \rangle. \quad (16)$$

The retained and complementary operators are therefore not independent phenomenological response functions. They are two faces of one positive load.

4.2 Positive master geometry

A closure geometry is built by positive Gram constructions rather than assumed as a free next-level metric:

$$\mathbf{H}_{\text{C}} := \mathbf{B}_{\text{C}}^\dagger \mathbf{W}_{\text{N}} \mathbf{B}_{\text{C}} \geq 0, \quad (17)$$

$$\mathbf{Y}_{\text{C}} := \mathbf{L}_{\text{C}}^\dagger \mathbf{H}_{\text{C}} \mathbf{L}_{\text{C}} \geq 0. \quad (18)$$

This form is stable under compression and supplies a nonnegative load to the next neutral resolvent. The graph metric associated with any such load is

$$\mathbf{G} := \mathbf{I} + \mathbf{Y} > 0, \quad \|\mathbf{z}\|_{\text{G}}^2 := \langle \mathbf{z}, \mathbf{G}\mathbf{z} \rangle. \quad (19)$$

For closed densely defined $\mathbf{K}$ on a separable Hilbert space, $\mathbf{Y} = \mathbf{K}^\dagger \mathbf{K}$ is nonnegative self-adjoint; $\mathbf{C}_{\text{N}}$ and $\mathbf{B}_{\text{N}}$ remain bounded by the spectral theorem, while $\mathbf{G}$ is interpreted through the closed quadratic-form norm $\|\mathbf{z}\|^2 + \|\mathbf{Y}^{\wedge \{1/2\}} \mathbf{z}\|^2$.

4.3 Exact neutral contrast representation

$$\mathbf{Z} := \mathbf{C}_{\text{N}}(\mathbf{Y}) - \mathbf{B}_{\text{N}}(\mathbf{Y}) = (\mathbf{I} - \mathbf{Y})(\mathbf{I} + \mathbf{Y})^{-1}. \quad (20)$$

$$\mathbf{C}_{\text{N}}(\mathbf{Y}) = \frac{1}{2}(\mathbf{I} + \mathbf{Z}), \quad \mathbf{B}_{\text{N}}(\mathbf{Y}) = \frac{1}{2}(\mathbf{I} - \mathbf{Z}). \quad (21)$$

Thus Z is a lossless one-operator encoding of the exact neutral pair. If $y \geq 0$ lies in $\sigma(Y)$, then $z = (1 - y)/(1 + y)$, so $-1 < z \leq 1$. The compact representation normalizes the neutral balance without introducing a new degree of freedom.

5. Recursive persistence and the canonical graph-metric projector

Recursive closure must distinguish structure that can remain present under repeated propagation from transient structure that decays or escapes. The selection rule must be intrinsic and must not use a target-aware spectral window.

$$R_C := M R M^\dagger, \quad M := \sqrt{N_R}, \quad (22)$$

where the detailed carrier realization of R and N_R may vary, but the persistence criterion below is stated directly in terms of R_C and the positive graph metric.

$$\mathcal{W} := \text{cl span}\{z : \sup_{k \geq 0} \|R_C^k z\|_G < \infty, \limsup_{k \rightarrow \infty} \|R_C^k z\|_G > 0\}. \quad (23)$$

In the finite-dimensional power-bounded case, $\mathcal{W}$ is the direct sum of the semisimple peripheral eigenspaces $|\lambda| = 1$. Let the remainder of the spectrum lie strictly inside the unit disk.

$$\gamma := 1 - \sup\{|\lambda| : \lambda \in \sigma(R_C) \setminus \sigma_{\text{per}}\} > 0. \quad (24)$$

When the peripheral block is isolated, the corresponding Riesz projector is a representation of the already-defined invariant subspace rather than a tunable selection.

$$P^R = (2\pi i)^{-1} \oint_{\Gamma} (zI - R_C)^{-1} dz, \quad \text{Ran}(P^R) = \mathcal{W}. \quad (25)$$

The unique orthogonal projector onto $\mathcal{W}$ in the graph metric is

$$P := W(W^\dagger G W)^{-1} W^\dagger G, \quad (26)$$

$$P^2 = P, \quad P^\dagger G = G P, \quad \|Pz\|_G \leq \|z\|_G. \quad (27)$$

Here the columns of W span $\mathcal{W}$. The expression is basis-independent. If the persistent spectrum is not isolated, or the bounded non-decay subspace is not well defined on the declared domain, reclosure fails rather than choosing a convenient contour.

6. Endogenous reciprocal order and the non-amplifying cross-fed packet

6.1 Persistent keep, upward, and export components

$$D^{\{\text{keep}\}} := P C_N D, \quad (28)$$

$$D^{\{\uparrow\}} := P B_N D, \quad (29)$$

$$D^{\{\text{out}\}} := (I - P) B_N D. \quad (30)$$

$$\Lambda^{\{\text{keep}\}} := \|D^{\{\text{keep}\}}\|_G^2, \quad \Lambda^{\{\uparrow\}} := \|D^{\{\uparrow\}}\|_G^2, \quad S := \Lambda^{\{\text{keep}\}} + \Lambda^{\{\uparrow\}}. \quad (31)$$

The first two loads measure the two complementary persistent channels. Export is not recursively cross-fed. A genuine dual reclosure requires both persistent loads to be strictly positive.

6.2 Reciprocal Balance Principle

The internal order pair is fixed by two equations: positive simplex normalization and equality of order-weighted closure loads.

$$\omega^{\{\text{keep}\}} + \omega^{\{\uparrow\}} = 1, \quad (32)$$

$$\omega^{\{\text{keep}\}} \Lambda^{\{\text{keep}\}} = \omega^{\{\uparrow\}} \Lambda^{\{\uparrow\}}. \quad (33)$$

$$\omega^{\{\text{keep}\}} = \Lambda^{\{\uparrow\}} / S, \quad \omega^{\{\uparrow\}} = \Lambda^{\{\text{keep}\}} / S. \quad (34)$$

$$r_U := \omega^{\{\uparrow\}} / \omega^{\{\text{keep}\}} = \Lambda^{\{\text{keep}\}} / \Lambda^{\{\uparrow\}}. \quad (35)$$

Theorem 2 — Reciprocal-order uniqueness. For $\Lambda^{\{\text{keep}\}} > 0$ and $\Lambda^{\{\uparrow\}} > 0$, simplex normalization and reciprocal balance have the unique positive solution in Equation (34). Hence the dual-order pair contains one projective degree of freedom fixed by the load ratio; it is not a pair of fitted parameters.

6.3 Cross-fed packet and exact local non-expansion

$$D^{\{\text{DO}\}} := \sqrt{\omega^{\{\text{keep}\}}} D^{\{\text{keep}\}} \oplus \sqrt{\omega^{\{\uparrow\}}} D^{\{\uparrow\}}. \quad (36)$$

$$\|D^{\{\text{DO}\}}\|_{G \oplus G}^2 = 2 \Lambda^{\{\text{keep}\}} \Lambda^{\{\uparrow\}} / S \leq S/2 \leq \|D\|_G^2 / 2. \quad (37)$$

Theorem 3 — Non-amplification of endogenous reclosure. Under the stated positivity, commutation, and graph-metric projection assumptions, the reciprocal cross-fed packet satisfies $\|\tilde{D}^{\wedge}\{DO\}\|_{G \oplus G} \leq \|D\|_G / \sqrt{2}$. Endogenous dual order therefore cannot act as a hidden structural gain source.

The first inequality is the arithmetic–geometric mean inequality. The second follows from graph-metric contractivity of P and $C_N^2 + B_N^2 \leq (C_N + B_N)^2 = I$ for commuting positive contractions.

7. Polar support transport and positive reconstruction

7.1 Canonical analysis operator

$$\mathcal{A}^{\wedge}\{DO\} := [\sqrt{\omega^{\wedge}\{\text{keep}\}} P C_N ; \sqrt{\omega^{\wedge}\{\uparrow\}} P B_N], \quad D^{\wedge}\{DO\} = \mathcal{A}^{\wedge}\{DO\} D. \quad (38)$$

The next-support transport is not postulated. It is the polar partial isometry of the already fixed analysis operator.

$$Q := (\mathcal{A}^{\wedge}\{DO\} \mathcal{A}^{\wedge}\{DO\})^{\wedge}\{1/2\}, \quad J := \mathcal{A}^{\wedge}\{DO\} Q^{\wedge}+. \quad (39)$$

$$J \dagger J = P_{\{\text{supp}(Q)\}}, \quad J J \dagger = P_{\{\text{cl Ran}(\mathcal{A}^{\wedge}\{DO\})\}}. \quad (40)$$

The Moore–Penrose inverse $Q^{\wedge}+$ fixes the support convention and zero action on the kernel. On the active support J is an isometry. No arbitrary kernel extension or free synthesis matrix remains.

7.2 Capacity transport

$$\mathcal{H}_+ := \text{cl Ran}(J). \quad (41)$$

$$\mathcal{A}_+ := P_+(\mathcal{A} \oplus \mathcal{A}) P_{\perp} \upharpoonright_{\{\mathcal{H}_+\}} \approx J \dagger (\mathcal{A} \oplus \mathcal{A}) J, \quad \mathcal{A} \in \{\mathcal{D}, \mathcal{C}, \mathcal{N}\}. \quad (42)$$

$$K_+ := \mathcal{C}_+ + \mathcal{N}_+ - \mathcal{D}_+. \quad (43)$$

Self-adjointness is preserved by compression; positivity is preserved for positive capacity operators. The next formation operator K_+ is therefore derived from the current state and the universal transport rule.

7.3 Positive Gram rebuild

$$H_+ := B_C[D^{\wedge}\{DO\}] \dagger W_N[D^{\wedge}\{DO\}] B_C[D^{\wedge}\{DO\}] \geq 0, \quad (44)$$

$$Y_+ := L_C[D^{\wedge}\{DO\}] \dagger H_+ L_C[D^{\wedge}\{DO\}] \geq 0. \quad (45)$$

The next neutral pair $C_N(Y_+)$, $B_N(Y_+)$ is consequently well defined whenever the declared domains exist. Positivity is regenerated constructively rather than assumed by induction.

8. Canonical emergence of a next structure

8.1 Universal formation gate

The reclosure becomes a new nonterminal formation only when the persistent dual channel exists, the persistence block is isolated, and the reconstructed formation operator has the same determinate negative-mode geometry as the primitive formation problem.

$$g_U(X) := \mathbb{1}_{\{\Lambda^{\wedge}\{\text{keep}\} \Lambda^{\wedge}\{\uparrow\} > 0, \gamma > 0, \lambda_{\min}(K_+) < 0, \text{mult}(\lambda_{\min}(K_+)) = 1, \Delta_+ > 0\}}. \quad (46)$$

If $g_U(X) = 0$, the branch terminates at $\perp$. If $g_U(X) = 1$, define on the transported real profile sector

$$\mathcal{A}_+^{\wedge}\{\text{form}\}(\delta) := \frac{1}{2} \langle \delta, K_+ \delta \rangle + \frac{1}{4} \|\delta\|^4. \quad (47)$$

$$\delta^{\wedge*}\{+, \pm\} = \pm \sqrt{(-\lambda_{\{+, 0\}})} e_{\{+, 0\}}. \quad (48)$$

Theorem 4 — Canonical next-structure formation. If the current state is admissible and $g_U(X) = 1$, then the reconstructed next formation functional $\mathcal{A}_+^{\wedge}\{\text{form}\}$ has exactly two nonzero global minima related by polarity. The next nonterminal candidate is therefore generated by the same formation law as the initial branch, with K_+ derived canonically from the current closure rather than supplied by a stage-specific rule.

Proof. The polar transport and compression determine the next capacities and hence K_+ . The gate guarantees a simple negative minimum eigenvalue with positive gap. Theorem 1 then applies verbatim on the transported real profile sector.

8.2 Formation, reproduction, and genuine novelty

A nonzero next minimum is a newly formed candidate, but mathematical novelty must not be inferred merely because the recursion index, carrier dimension, or active support changed. Let $V_Q: \mathcal{H}_Q \rightarrow \mathcal{H}$ be any isometry with $\text{Ran}(V_Q) = \text{supp}(Q)$, where $Q = (\mathcal{A}^{\wedge}\{DO\} \mathcal{A}^{\wedge}\{DO\})^{\wedge}\{1/2\}$. The canonically inherited active-support reference $\Xi_{\text{inh}}(X)$ is defined as the simultaneous structure-preserving isomorphism class of the predecessor data compressed by V_Q to $\mathcal{H}_Q$, including in particular $K_{\text{inh}} := V_Q \dagger K V_Q$, $Y_{\text{inh}} := V_Q \dagger Y V_Q$, the compressed neutral and persistence data,

the support rank, and the scalar reciprocal-order invariants. $\Xi_{\text{inh}}(X)$ is a reference signature, not a second generator state and not a re-neutralized fit. Different choices of V_Q differ only by a unitary change of basis on $\mathcal{H}_Q$, so the class $\Xi_{\text{inh}}(X)$ is basis independent. Let $\Xi(X_+)$ denote the corresponding canonical signature of the actually reconstructed next state.

$$\text{Novel_strict}(X \rightarrow X_+) = 1 \Leftrightarrow \Xi(X_+) \neq_{\text{can}} \Xi_{\text{inh}}(X). \quad (49)$$

If $g_U=1$ but $\Xi(X_+) \neq_{\text{can}} \Xi_{\text{inh}}(X)$, reclosure represents stable inheritance, reproduction, or persistence of the same structural type on the active support. If $g_U=1$ and the signatures are inequivalent, a new canonical structural class has emerged. This criterion blocks the false positive in which the full predecessor and the next state differ only because transient directions were discarded. A difference in any canonical invariant is sufficient to establish inequivalence; in particular, $\sigma(K_+) \neq \sigma(K_{\text{inh}})$ is a sufficient spectral witness of novelty. Equality of formation spectra alone is not sufficient to prove full canonical equivalence.

Figure 1. Canonical generation cycle for a new structure.

Current closure X	Exact neutral resolution	Persistent projection	Reciprocal order	Polar support transport	Positive Gram rebuild	Formation gate	Next closure X_+ or $\perp$
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9. Compact universal reclosure equation and exact equivalence

9.1 Compact split operator

Using the neutral contrast Z , define the reciprocal-order block operator

$$\Omega := \text{diag}(\sqrt{\omega}^{\{\text{keep}\}} I, \sqrt{\omega}^{\{\uparrow\}} I). \quad (50)$$

$$S_X := \frac{1}{2} \Omega [P(I+Z); P(I-Z)]. \quad (51)$$

$$S_{XD} = D^{\{\text{DO}\}}. \quad (52)$$

Thus the compact split is algebraically identical to the expanded persistent keep/upward construction.

9.2 Canonical reclosure functional

Let $\mathfrak{R}_X$ denote only the ordered composition of already-defined operations after the split: polar support decomposition, capacity compression, positive Gram reconstruction, exact formation gate, determinate formation, and final neutral reclosure. $\mathfrak{R}_X$ is a composition symbol, not a new free map.

$$U[X] := \mathfrak{R}_X(S_{XD}[X]), \quad (53)$$

$$X_{\{n+1\}} = U[X_n], \quad X_n = U^{\wedge n}[X_0], \quad U(\perp) = \perp. \quad (54)$$

Theorem 5 — Compact/expanded equivalence. For every admissible state on which the expanded operators are defined, the compact Equations (20)–(52) produce exactly the same cross-fed packet, transport support, reconstructed geometry, gate decision, and next state as the expanded audit construction.

Proof. Equations (21) recover C_N and B_N exactly from Z . Substitution into S_X gives $\sqrt{\omega}^{\{\text{keep}\}} PC_{ND} \oplus \sqrt{\omega}^{\{\uparrow\}} PB_{ND} = \tilde{D}^{\{\text{DO}\}}$. All downstream steps are identical by definition of $\mathfrak{R}_X$.

10. Totalization, self-closure, and no-hidden-retuning

The universal operator is total on $\bar{X}_{\text{adm}}$ because every failed gate returns $\perp$ and $\perp$ is absorbing. The stronger question is whether every nonterminal output carries exactly the same kinds of objects required by the next application.

Theorem 6 — Canonical self-closure. Assume (i) determinate formation; (ii) exact neutral elimination with $Y \geq 0$; (iii) a well-defined isolated bounded non-decay persistent sector; (iv) graph-metric orthogonal projection; (v) strictly positive dual persistent loads; (vi) reciprocal balance and simplex normalization; (vii) support polar transport; (viii) positive Gram reconstruction; and (ix) the common exact formation gate. Then $U: \bar{X}_{\text{adm}} \rightarrow \bar{X}_{\text{adm}}$ is a totalized level-independent self-map. Every nonterminal output regenerates the positive load, neutral pair, persistence data, reciprocal order, transported capacities, and formation operator needed for the next iteration. No stage label, external order scalar, selectable contour, free kernel extension, fitted transport matrix, or numerical threshold is part of the generator.

The theorem is deliberately conditional. Its value is that every internal choice surface is either fixed by an exact construction or exposed as a failure condition.

Table 3. Freedom-closure ledger for the canonical generator.

Potential freedom	Canonical closure rule	Residual status
External order parameter	Reciprocal balance fixes ω from persistent loads.	None in $\mathcal{U}$.
Persistence window	Use bounded non-decay; Riesz contour only represents an already isolated block.	No tunable window.
Projection metric	$G=I+Y$ fixes the graph-metric orthogonal projector.	Unique.
Kernel extension	Moore–Penrose support polar factor with zero kernel action.	Unique.
Next-support transport	Polar partial isometry of $\mathcal{A}^{\{DO\}}$.	Unique on active support.
Next geometry	Positive Gram rebuild from transported packet.	Derived from X .
Formation choice	Simple negative mode + positive gap; degeneracy terminates.	No tie-break.
Stage label	Absent from generator.	Downstream realization only.
Mathematical tolerance	Exact theorem conditions.	Measurement layer only.

11. Intrinsic behavioral quotient and structural novelty

11.1 Complete canonical data stream

To prevent target-dependent coarse-graining, define $\Xi_U(X)$ as the complete canonical datum generated by one application of the frozen operator grammar. Its future behavioral signature is the sequence of isomorphism classes along the entire orbit.

$$\mathfrak{B}_U(X) := ([\Xi_U(\mathcal{U}^n X)]_{\text{iso}})_{n \geq 0}. \quad (55)$$

$$X \equiv_U X' \Leftrightarrow \mathfrak{B}_U(X) = \mathfrak{B}_U(X'). \quad (56)$$

The relation is automatically a $\mathcal{U}$ -congruence because equality of the full future stream is preserved by dropping the first term.

$$Q_U := X_{\text{adm}} / \equiv_U, \quad \bar{U}_U([X]) := [U(X)]. \quad (57)$$

Theorem 7 — Intrinsic behavioral quotient. The quotient Q_U and induced map $\bar{U}_U$ are well defined without choosing a target-dependent stratum or congruence. Any mature theory or natural carrier can enter only after this quotient has been fixed.

11.2 Strong novelty in the quotient

$$\text{StrongNovel}(X \rightarrow X_+) = 1 \Leftrightarrow \text{Novel_strict}(X \rightarrow X_+) = 1 \wedge [X_+]_{\equiv_U} \neq [X]_{\equiv_U}. \quad (58)$$

This criterion requires both strict structural novelty relative to the canonically inherited active-support reference and a difference in complete future canonical behavior relative to the predecessor orbit. The first conjunct prevents projection or support reduction from being mistaken for emergence; the second excludes merely instantaneous representational differences that collapse into the same recursive organization.

11.3 Finite quotient stabilization

Theorem 8 — Finite quotient stabilization. If Q_U is finite, the descending image chain $Q_U \supseteq \bar{U}_U(Q_U) \supseteq \bar{U}_U^2(Q_U) \supseteq \dots$ stabilizes after finitely many steps. On the stabilized image, $\bar{U}_U$ is surjective and therefore bijective.

Proof. A descending chain of finite subsets cannot shrink indefinitely. Once $I_m = I_{m+1}$, $\bar{U}_U$ maps I_m onto itself. Surjectivity of a self-map on a finite set implies injectivity.

This gives an exact finite analogue of transient-to-persistent closure: many-to-one reduction may feed a recursively persistent sector without requiring an inverse map for discarded distinctions.

12. No-choice theory reduction and realization boundary

12.1 Exact law recovery

For a candidate target theory $T_\lambda = (S_T, F_{\{T, \lambda\}}, \mathcal{O}_T)$, define the canonical orbit closure

$$\mathcal{C}_U([X]) := \text{cl}\{\bar{U}_U^n([X]) : n \in \mathbb{N}\}. \quad (59)$$

Exact recovery is tested in one recursion step:

$$\Psi \circ \bar{U}_U = F_{\{T, \lambda\}} \circ \Psi \text{ on } \mathcal{C}_U([X]). \quad (60)$$

$$\Theta_T([X]) := \{\lambda : \exists \Psi \text{ satisfying exact recovery and canonical observable preservation}\}. \quad (61)$$

$$\Theta_T([X]) / \sim_{\{\text{units}\}} = \{[\lambda_U]\}. \quad (62)$$

The representation witness Ψ is not a tunable generator object. Witnesses differing only by automorphisms commuting with the target dynamics are representation gauge. If several physically inequivalent parameter classes survive, parameter recovery is incomplete and no preferred value may be selected.

12.2 Five-level recovery certificate

Table 4. Five-level no-choice recovery certificate.

Level	Pass condition	Failure interpretation
L1 Endogenous branch	The candidate branch lies in an orbit closure already generated in $\mathcal{Q}_U$.	Only analogy.
L2 Intrinsic quotient	$\equiv_U$ and $\bar{U}_U$ were fixed before naming the target.	Target-specific coarse-graining would be required.
L3 Exact law recovery	One-step exact conjugacy or declared exact semiconjugacy.	Known law is being imported or only approximated.
L4 Parameter/observable closure	One physical parameter class modulo units; target observables pull back canonically.	Hidden parameter/readout freedom remains.
L5 Independent realization	Measured carrier transitions commute with the frozen quotient dynamics under preregistered measurement uncertainty.	Mathematics not yet shown to describe the natural carrier.

The certificate prevents mathematical closure from being presented as empirical confirmation. L5 is intentionally external to the generator.

13. Failure surfaces, falsification, and preregistration

A closed mathematical architecture is scientifically useful only if failed realizations cannot be repaired by silently changing its internal rules. The following failures are therefore explicit and non-negotiable.

Table 5. Primary mathematical and empirical failure surfaces.

Failure surface	Decision condition
Formation failure	$\lambda_{\min}(K) \geq 0$, non-simple minimum, or no positive spectral gap where determinate formation is claimed.
Neutral failure	The required response cannot be represented by the exact complementary resolvent pair without changing the generator.
Persistence failure	Bounded non-decay is ill-defined or the peripheral block is not isolated under the stated assumptions.
Dual-order failure	A persistent load vanishes, reciprocal balance is violated, or an external order coefficient is required.
Transport failure	The support polar factor is not well defined on the declared domain or requires a target-specific extension.
Positivity failure	The Gram rebuild does not yield a nonnegative next load

Failure surface	Decision condition
	under the stated construction.
Generativity failure	The next formation gate fails; branch maps to $\perp$.
Novelty failure	The next canonical signature is isomorphic to the canonically inherited active-support reference $\Xi_{\text{inh}}(X)$ when a new structural class is claimed.
Reduction failure	Exact law recovery leaves multiple inequivalent parameter classes or requires a free observable bridge.
Realization failure	Independent carrier transitions do not commute with frozen quotient dynamics.

For empirical work, the carrier-to-BFG role map, numerical estimator, measurement uncertainty model, quotient coordinates, and null models must be frozen before outcome inspection. Mathematical tolerances do not enter the exact generator; only empirical measurement models may carry tolerances.

14. Independent numerical consistency audit

Two numerical audits were performed. First, 10,000 seeded complex random consistency instances with dimensions $d=2,\dots,8$ tested the exact neutral pair, graph-metric projection, reciprocal-order weights, compact/expanded split, SVD polar partial isometry, positive compression, and positive Gram rebuild; 1,000 random real formation operators with one simple negative eigenvalue tested formation stationarity. Second, an independent 10,000-trial emergence ensemble reconstructed K_+ and compared its formation spectrum with the canonically inherited support compression K_{inh} . Of these trials, 9,993 passed the next-formation gate and 7 terminated. Among the successful trials, the normalized spectral novelty witness $\|\text{sort } \sigma(K_+) - \text{sort } \sigma(K_{\text{inh}})\|_2 / \|\text{sort } \sigma(K_{\text{inh}})\|_2$ ranged from 0.0362 to 0.8248 with median 0.2698. A separate set of 1,000 commuting controls, in which K , Y , the persistent projector, and the neutral pair shared one eigenbasis, yielded a maximum normalized K -spectrum mismatch of 7.8×10^{-16} . This control isolates mere active-support inheritance: it does not by itself prove equality of the full canonical signature, but it shows that the spectral novelty witness does not fire merely because support compression or recursion occurred.

Table 6. Independent numerical audit of the consolidated canonical equations.

Audit quantity	Observed result	Exact expectation
Neutral partition max $\ C_N + B_N - I\ _2$	7.16×10^{-16}	0
Reciprocal balance max normalized residual	1.19×10^{-16}	0
Largest observed cross-fed gain	0.624982	$\leq 1/\sqrt{2} \approx 0.707107$
Compact/expanded packet max relative error	3.70×10^{-14}	0
Polar support max $\ J^\dagger J - P_{\text{supp}}\ _2$	3.45×10^{-15}	0
Smallest eigenvalue of compressed positive operator	-8.99×10^{-16}	≥ 0 up to roundoff
Smallest eigenvalue of rebuilt Gram load	1.86×10^{-9}	≥ 0
Max formation-stationarity residual	6.00×10^{-15}	0
Emergence gate passes	9,993 / 10,000	Diagnostic outcome; failures terminate
Successful trials with nonzero spectral novelty witness	9,993 / 9,993	$\sigma(K_+) \neq \sigma(K_{\text{inh}})$ is sufficient for inequivalence
Spectral novelty witness, min / median / max	0.0362 / 0.2698 / 0.8248	Diagnostic, not a universal constant

Audit quantity	Observed result	Exact expectation
Commuting inheritance control max spectral mismatch	7.8×10^{-16}	0 up to roundoff
Corr(spectral novelty, normalized $\ KP - PK\ $)	0.781	Diagnostic association only

Interpretation: These are finite-precision consistency and discrimination checks of the mathematical construction. The emergence ensemble shows that the reconstructed formation operator can acquire invariant spectral data not present in the no-mixing inherited support reference, while the commuting control preserves the inherited formation spectrum to roundoff. In the noncommuting ensemble, the normalized formation-spectrum novelty witness correlated with the normalized commutator $\|KP - PK\|$ at $r \approx 0.781$. This identifies noncommutative persistence selection as a strong diagnostic of the tested novelty mechanism, not as a proved necessary condition. None of these calculations provides empirical evidence that a natural system realizes the framework.

15. Limitations and open proof obligations

The present paper is mathematically closed only relative to its declared category and assumptions. Several questions remain outside the proved core and should not be collapsed into it.

- The finite-dimensional real formation theorem is exact; extension of determinate symmetry breaking to infinite-dimensional profile spaces requires domain, compactness, and bifurcation control beyond the present statement.
- The bounded non-decay persistence definition is canonical, but spectral isolation can fail for continuous spectrum or non-normal operators; such cases require a carefully stated generalized persistence theory rather than an ad hoc contour.
- The graph-metric projector is unique for closed subspaces in the declared Hilbert geometry; unbounded form domains require closed-form compatibility.
- Polar support transport is canonical on the active support. Whether a particular carrier interpretation assigns all physically relevant structure to that support is an empirical or model-specific question.
- The reclosure theorem establishes generation of admissible next candidates, not that every successful gate produces a genuinely new structural class and not that every natural hierarchy follows the same orbit. Strict novelty requires inequivalence to the canonically inherited active-support reference. In numerical work, a changed formation spectrum is a sufficient witness of this inequivalence, whereas an unchanged spectrum does not by itself establish full canonical equivalence because non-spectral invariants may still differ.
- The intrinsic behavioral quotient is mathematically well defined at the formal level, but computing complete future data may be infeasible; practical finite-horizon approximations must not be confused with the exact equivalence.
- Exact reduction of established theories is a proof obligation. Compatibility, analogy, or qualitative resemblance is not exact recovery.
- The framework does not identify structural closure loads with joule-valued physical energy without an independent carrier-level derivation.

16. Conclusion

The canonical universal reclosure architecture can be stated without developmental version labels and without retaining the incompatible discretionary freedoms of earlier formulations. Its mathematical spine is compact: a positive load yields an exact complementary neutral pair; bounded non-decay selects what can persist; graph-metric projection isolates the persistent channels; reciprocal balance fixes their endogenous order weights; the cross-fed packet is non-amplifying; polar decomposition fixes next-support transport; positive Gram reconstruction regenerates the geometry; and the same negative-mode formation law either produces the next determinate structure or terminates the branch at $\perp$.

The central emergence claim is therefore precise. New structure is not inserted by naming a stage and is not inferred from recursion depth or dimension change. It appears first as a determinate next candidate when the canonically reconstructed formation operator crosses the formation gate and its quartic admissibility functional develops a nonzero polarity pair. A genuinely new structural class is claimed only when the next canonical signature is inequivalent to the canonically inherited active-support reference of the predecessor. If the next signature is isomorphic to that inherited reference, the recursion describes persistence, reproduction, or support

reduction. Strong novelty additionally requires a different future behavioral class in the intrinsic quotient. This makes formation, inheritance, structural novelty, recursive novelty, and failure separately auditable.

The compact law $\mathbb{U}[X]=\mathfrak{R}_X(\mathcal{S}_X\mathcal{D}[X])$ is only the display form. Every symbol expands uniquely into exact neutral resolution, intrinsic persistence, reciprocal order, polar support transport, positive rebuild, and determinate formation. The intrinsic future-behavioral quotient then removes target-dependent coarse-graining and forces mature-theory recovery to succeed without fitted iterate multipliers, arbitrary parameter selection, or free observable bridges. Mathematical closure is thereby made as strong as the assumptions permit, while empirical natural realization remains openly falsifiable.

Appendix A. Proof register

A.1 Determinate formation

For $\delta = ru$ with $r \geq 0$ and $\|u\| = 1$, Rayleigh–Ritz gives $\langle u, Ku \rangle \geq \lambda_0$. Therefore $\mathcal{A}_{\text{Adm}}(ru) \geq \frac{1}{2}\lambda_0 r^2 + \frac{1}{4}r^4$. If $\lambda_0 \geq 0$, the right side is minimized only at $r = 0$. If $\lambda_0 < 0$, its derivative is $r(\lambda_0 + r^2)$, so the nonzero minimizing radius is $r = \sqrt{-\lambda_0}$. Equality in Rayleigh–Ritz requires u to lie in the lowest eigenspace. Simplicity gives $u = \pm e_0$. The second radial derivative at the nonzero radius is $-2\lambda_0 > 0$, establishing strict radial stability; the positive spectral gap controls transverse stability.

A.2 Reciprocal-order uniqueness

With positive loads $a = \Lambda^{\uparrow\text{keep}}$, $b = \Lambda^{\uparrow}$, solve $x + y = 1$ and $xa = yb$. Substitution $y = 1 - x$ gives $x(a + b) = b$, hence $x = b/(a + b)$ and $y = a/(a + b)$. Positivity and uniqueness follow immediately.

A.3 Non-expansion

The cross-fed norm is $\omega^{\text{keep}}\Lambda^{\text{keep}} + \omega^{\uparrow}\Lambda^{\uparrow} = 2ab/(a + b) \leq (a + b)/2$ by $4ab \leq (a + b)^2$. Because P is G -contractive and C_N, B_N are commuting positive contractions with $C_N + B_N = I$, one has $C_N^2 + B_N^2 \leq I$, giving $a + b \leq \|D\|_G^2$ under the declared commutation/metric assumptions.

A.4 Compact/expanded equivalence

From $Z = C_N - B_N$ and $C_N + B_N = I$, solve $C_N = (I + Z)/2$ and $B_N = (I - Z)/2$. Substitution into S_X gives exactly $\sqrt{\omega^{\text{keep}}}\text{PC}_{\text{ND}} \oplus \sqrt{\omega^{\uparrow}}\text{PB}_{\text{ND}}$. Since all remaining operations are the same ordered composition, the compact and expanded maps coincide.

A.5 Finite quotient stabilization

Let $I_n = \bar{U}_U^n(Q_U)$. Then $I_{n+1} \subseteq I_n$. If Q_U is finite, cardinality can decrease only finitely many times, so $I_m = I_{m+1}$ for some m . The restriction $\bar{U}_U: I_m \rightarrow I_m$ is surjective by definition of I_{m+1} ; any surjection of a finite set onto itself is bijective.

A.6 Basis independence of the inherited active-support reference

If V_Q and V'_Q are two isometries onto $\text{supp}(Q)$, then $V'_Q = V_Q U$ for a unitary U on $\mathcal{H}_Q$. For every predecessor operator A included in the inherited signature, $V'_Q \dagger A V'_Q = U \dagger (V_Q \dagger A V_Q) U$. Therefore spectra, ranks, positivity, algebraic relations, and all simultaneous unitary-isomorphism invariants of the compressed family are unchanged. The reference $\Xi_{\text{inh}}(X)$ is consequently independent of the arbitrary basis used to represent the active support. This is precisely why comparison with $\Xi_{\text{inh}}(X)$, rather than with the full predecessor state, separates structural transformation from mere deletion of transient directions.

Appendix B. Core formula register

Table 7. Canonical formula register.

Object	Canonical expression	Location
Formation operator	$K = c + \kappa - \mathcal{Q}$	Eq. (5)
Admissibility functional	$\mathcal{A}_{\text{Adm}} = \frac{1}{2} \langle \delta, K \delta \rangle + \frac{1}{4} \ \delta\ ^4$	Eq. (6)
Neutral load	$Y = A \dagger A \geq 0$	Eq. (10)
Neutral pair	$C_N = (I + Y)^{-1}$; $B_N = Y(I + Y)^{-1}$	Eqs. (12)–(13)
Neutral contrast	$Z = (I - Y)(I + Y)^{-1}$	Eq. (20)
Graph metric	$G = I + Y$	Eq. (19)
Persistent witness	bounded non-decaying subspace $\mathcal{W}$	Eq. (23)
Persistent projector	$P = W(W \dagger G W)^{-1} W \dagger G$	Eq. (26)
Reciprocal order	$\omega_{\text{keep}} = \Lambda_{\uparrow} / S$; $\omega_{\uparrow} = \Lambda_{\text{keep}} / S$	Eq. (34)
Cross-fed packet	$\tilde{D}_{\text{DO}} = \sqrt{\omega_{\text{keep}}} D_{\text{keep}} \oplus \sqrt{\omega_{\uparrow}} D_{\uparrow}$	Eq. (36)
Polar transport	$Q = (\mathcal{A} \dagger \mathcal{A})^{\wedge \{1/2\}}$; $J = \mathcal{A} Q^{\wedge +}$	Eq. (39)
Next formation operator	$K_{++} = c_{++} + \kappa_{++} - \mathcal{Q}_{++}$	Eq. (43)

Object	Canonical expression	Location
Formation gate	dual loads + spectral gap + simple negative next mode	Eq. (46)
Compact split	$S_X = \frac{1}{2}\Omega[P(I+Z); P(I-Z)]$	Eq. (51)
Universal map	$U[X] = \Re_X(S_{XD}[X])$	Eq. (53)
Behavioral quotient	$Q_U = \bar{X}_{adm} \equiv U$	Eq. (57)
Inherited active-support reference	$\Xi_{inh}(X)$: canonical class of predecessor data compressed by V_Q onto $\text{supp}(Q)$	Sec. 8.2
Strict novelty	$\Xi(X_+) \neq_{can} \Xi_{inh}(X)$	Eq. (49)

Appendix C. Reproducibility register

Table 8. Numerical audit reproducibility register.

Item	Specification
Random seed	20260922
Complex operator trials	10,000
Dimensions	Uniform integer $d \in \{2, \dots, 8\}$
Positive load	$Y = A^\dagger A$ with complex Gaussian A scaled by dimension
Graph metric	$G = I + Y$
Persistent projector	Random full-rank W , $P = W(W^\dagger GW)^{-1}W^\dagger G$
Neutral pair	$C_N = (I + Y)^{-1}$; $B_N = Y(I + Y)^{-1}$
Reciprocal weights	$\omega_{keep} = \Lambda_{\uparrow} / (\Lambda_{keep} + \Lambda_{\uparrow})$; $\omega_{\uparrow} = \Lambda_{keep} / (\Lambda_{keep} + \Lambda_{\uparrow})$
Polar factor	SVD implementation of support partial isometry
Gram test	Random positive W_N and $H = B^\dagger W_{NB}$; $Y_+ = L^\dagger HL$
Formation test	1,000 random real symmetric K with one simple negative eigenvalue
Audit interpretation	Internal finite-precision consistency only; no empirical dataset
Emergence audit trials	10,000 independent reconstructed next-formation trials; 9,993 passed the formation gate.
Formation-spectrum novelty witness	$\ \text{sort } \sigma(K_+) - \text{sort } \sigma(K_{inh})\ _2 / \ \text{sort } \sigma(K_{inh})\ _2$ on successful trials.
Commuting inheritance controls	1,000 controls with K , Y , P , C_N and B_N sharing a common eigenbasis; compare K_+ with K_{inh} on active support.
Commuting-control result	Maximum normalized formation-spectrum mismatch 7.8×10^{-16} .
Noncommutativity diagnostic	Pearson correlation between normalized spectral novelty and normalized $\ KP - PK\ $: $r \approx 0.781$ in the emergence ensemble.

References

- [1] M. T. Wende, “BFG Universal Structural Whitepaper — Strong Form: Local Physical Closure Edition,” Balance-Field Framework, independent research whitepaper, September 2026.

- [2] M. T. Wende, “Balance–Field Theory: Universal Structural Whitepaper — Reviewed, Corrected, Equation-Numbered, and Literature-Referenced Edition,” Balance–Field Framework, September 2026.
- [3] M. T. Wende and T. W. Loker, “A Constructive Level-0 Foundation of the Balance-Field Framework (BFG): A Pre-Mathematical Balance Substrate for Emergent Order, Information, Process Time and Physics,” internal BFG manuscript, 11 March 2026.
- [4] M. T. Wende and L. K. C. Leighton, “Recursive Conjugate-Closure Extension of the Balance–Field Framework: $R_C = \text{MRM}^\dagger$ and $M = \sqrt{N}$,” Balance–Field Framework manuscript, 2026.
- [5] M. T. Wende, “The Order-Indexed Neutral Interface: Recursive Closure Networks, Abiogenesis, and Biological Self-Unfolding,” Balance–Field Framework theoretical whitepaper, 1 September 2026.
- [6] R. A. Horn and C. R. Johnson, Matrix Analysis, 2nd ed., Cambridge University Press, 2013. doi:10.1017/CBO9781139020411.
- [7] C. Davis and W. M. Kahan, “The Rotation of Eigenvectors by a Perturbation. III,” SIAM Journal on Numerical Analysis, 7, 1–46 (1970). doi:10.1137/0707001.
- [8] T. Kato, Perturbation Theory for Linear Operators, 2nd ed., Springer, 1995. doi:10.1007/978-3-642-66282-9.
- [9] M. Reed and B. Simon, Methods of Modern Mathematical Physics, Vol. I: Functional Analysis, revised ed., Academic Press, 1980.
- [10] J. Jachymski, I. Jóźwik, and M. Terepeta, “The Banach Fixed Point Theorem: Selected Topics from Its Hundred-Year History,” RACSAM 118, 140 (2024). doi:10.1007/s13398-024-01636-6.

Declarations

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Author contribution. Conceptualization, consolidation of the universal reclosure architecture, mathematical formulation, interpretation, and manuscript responsibility: Marcel Theodor Wende. Earlier Level-0 and recursive conjugate-closure source lines are credited in the bibliography.

Data availability. No external empirical dataset was analyzed.

Code and numerical audit. The random seed, dimensions, operator constructions, and reported numerical diagnostics are stated in Appendix C. The audit tests internal mathematical consistency only.

Ethics. No human participants, animal experiments, or identifiable personal data were used.

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